Yi Zhan

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Research Interests

- Fluid-structure interactions (FSI) in coastal engineering
- High-performance computing for Smoothed Particle Hydrodynamics (SPH)
- Multi-agent deep reinforcement learning (MADRL) for fluid mechanics and flow control

Education

University of Vigo, Visiting Scholar at EphysLab (Spain) 04/2025 – 04/2026

- Advisor: Alejandro J. C. Crespo [Link]

Zhejiang University, PhD in Marine Technology and Engineering (China) 09/2020 – 09/2026

- Advisor: Min Luo [Link], Abbas Khayyer [Link]

Hohai University, BS in Harbour, Coastal and Offshore Engineering (China) 09/2016 – 06/2020

- GPA: 4.44/5.0, ranking 4/18
- Thesis: Developed a two-dimensional numerical model for tidal wave simulation

Research Experience

Hydrodynamic characteristics analysis and optimization of multi-connected floating flexible structures 06/2025 – Present

- Develops a coupled fluid-structure interaction model for multiple hinged flexible floating bodies simulations based on DualSPHysics+
- The model is validated through physical experiments, the motion, deformation response and hydrodynamic characteristics of the array under extreme wave conditions are investigated

Develops GPU-parallelized simulation platform coupling MADRL and SPH models 09/2024 – 12/2025

- Integrates the SPH solver DualSPHysics+ with machine learning framework LibTorch, enabling in-time communications between high-fidelity fluid simulations and intelligent control agents
- Applies the SPH-MADRL model to wave-energy point-absorber array, where active control of the power take-off damping coefficient led to an overall 25% increase in energy capture efficiency

Enhances the SPH model for simulating fluid flow, structure deformation and fluid–structure interactions 09/2022 – 09/2024

- Proposes several novel numerical schemes to enhance the accuracy, stability, and energy conservation of the SPH method in free-surface flow simulations, while simultaneously reducing stress noise and suppressing hourglass modes in structural dynamics
- All the advanced schemes were implemented with GPU parallelization and incorporated into the open-source SPH framework DualSPHysics, leading to the development of DualSPHysics+

Publications

A. First-Author Publications

[A5] **Yi Zhan**, Iván Martínez-Estévez, Min Luo, Alejandro J.C. Crespo, and Abbas Khayyer. "Dynamic responses of flexible floating multi-module arrays under solitary waves: SPH simulation and experimental study." *Coastal Engineering*, submitted.

[A4] **Yi Zhan**, Iván Martínez-Estévez, Min Luo, Alejandro J.C. Crespo, and Abbas Khayyer. "Coupling Smoothed Particle Hydrodynamics with multi-agent deep reinforcement learning for cooperative control of point absorbers." *Engineering Applications of Artificial Intelligence*, 181 (2026): 115417. [Link]

[A3] **Yi Zhan**, Min Luo, and Abbas Khayyer. "DualSPHysics+: An enhanced DualSPHysics with improvements in accuracy, energy conservation and resolution of the continuity equation." *Computer Physics Communications*, 306 (2025): 109389. [Link]

[A2] **Yi Zhan**, Min Luo, and Abbas Khayyer. "An enhanced SPH-based hydroelastic FSI solver with structural dynamic hourglass control." *Journal of Fluids and Structures*, 135 (2025): 104295. [Link]

[A1] **Yi Zhan**, Min Luo, and Abbas Khayyer. "An enhanced numerical wave tank for wave–structure interaction based on DualSPHysics+." *Ocean Engineering*, 340 (2025): 122413. [Link]

B. Co-Author Publications

[B3] Guozhen Cai, Min Luo, Matteo Rubinato, **Yi Zhan**, and Abbas Khayyer. "Simulation of multi-body floating structures under wave actions using DualSPHysics+." *Ocean Engineering*, 349 (2026): 124073. [Link]

[B2] Xiujia Su, Chen Wang, Min Luo, and **Yi Zhan**. "Development of a smoothed particle hydrodynamics model for porous media flows with enhanced volume conservation and the revisit of the mass conservation equation." *Physics of Fluids*, 36 (2024). [Link]

[B1] Zhouteng Ye, Mark Sussman, **Yi Zhan**, and Xizeng Zhao. "A decision-tree based moment-of-fluid (DTMOF) method in 3D rectangular hexahedrons." *arXiv preprint*, arXiv:2108.02533 (2021). [Link]

Presentations

Coupling SPH with a Multi-Agent DRL Framework for Active Flow Control [Link] — 10/2025

- *Particles 2025*, Barcelona — Oral Presentation

An Enhanced SPH-Based FSI Solver with Dynamic Hourglass Control [Link] — 06/2025

- *SPHERIC*, Barcelona — Oral Presentation

An enhanced DualSPHysics with improvements in accuracy, energy conservation and resolution of the continuity equation [Link] — 10/2024

- *SPHERIC*, Zhuhai — Oral Presentation, Outstanding Student Paper Award

Teaching Experience

Teaching Assistant, Computational fluid dynamics course, Zhejiang University — Fall 2023

- Lectured on the mesh-free particle method; designed and graded homework assignments

Relevant Skills

Programming Languages: C++, Cuda, Python, MATLAB

Software & Tools: DualSPHysics+, OpenFOAM, Paraview, PyTorch, LaTeX

Languages: Chinese (Native), English (Fluent in academic and daily communication)

Hobbies: Traveling, Badminton

Awards and Honors

National Scholarship for Joint-Training Program, China Scholarship Council — 2024

SPHERIC-2024 Excellent Student Paper Award, SPHERIC Committee — 2024

Award of Honor for Graduate, Zhejiang University — 2024

Excellent Graduate Student Award, Zhejiang University — 2023

Outstanding Student Award, Hohai University — 2020

National Scholarship, Ministry of Education of China — 2019

Third Prize, National College Student Mathematics Competition, Chinese Mathematical Society — 2018

Second Prize, Jiangsu Provincial Advanced Mathematics Competition, Jiangsu Society of Higher Mathematics — 2018